

Black Holes as Almost-Perfect All:All Entanglement Graphs: Interior Collapse, Horizon Wiring, and the Micro-Hole Phase Transition

Draft v0.6 — computational companion

`src/bh_graph` + `scripts/generate_figures.py` reproduce all figures (67 tests)

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Abstract

We formalize the intuition that a black hole interior is an almost-perfect all:all entanglement graph. Internal edges cost no exterior space; each of the k exterior legs costs $\sim l_p^2$ of horizon area, so $A(k) = k l_p^2$ independent of interior node count N . We show (1) the complete graph K_N destroys interior distance and, under finite-speed random circuits, derives the Sekino–Susskind $t_* \sim \log N$ fast-scrambling bound; (2) mass enters via a derived $k^*(N) = 16\pi(\varepsilon N/l_p)^2$ fixed point, spin/charge consume part of the exterior budget (extremal Kerr keeps half the legs), and the $k \rightarrow 0$ monogamy limit is a baby-universe pinch-off with an explicit nonlinear CKW frontier; (3) micro-holes are pointlike until $k l_p^2 \geq 4\pi r_{\text{point}}^2$, then pop a horizon via a computed QES/island crossing, with an LQG-style area gap. Evaporation as leg surgery reproduces the exact Page curve (Page 1993, $-1/2$ nat correction included) with Haar-typical fluctuations, and Hayden–Preskill mirror recovery works past $k \sim N/2$. Simulations in `bh_graph` accompany each claim.

1 Sec 1: All:all = no interior space

Interior wiring G_N compared across K_N , chain, grid, random regular under SI spread (cover time = seed eccentricity). Only K_N has diameter 1 and cover time 1 at every N (Figs. 1–2). Finite-speed random-matching circuits (Appendix A) promote this to a derived $t_* \approx \log_2 N / \log_2(1 + p)$.

family	diameter	t_{cover} (SI)	t_* (circuit)	gap
K_N	1	1	$\log_2 N$	N
chain	$N-1$	$\sim N/2$	$\sim N/v$	$\sim \pi^2/N^2$
grid	$\sim 2\sqrt{N}$	$\sim 2\sqrt{N}$	$O(\sqrt{N})$	$O(1/N)$
random-regular	$\sim \log N$	$\sim \log N$	$\sim \log N$	$O(1)$

Table 1: Scrambling hierarchy; `bh_graph.scrambling`, `circuits`.

2 Sec 2: Horizon area counts exterior legs

Interior pairs $N(N-1)/2$ plus $k \ll N^2$ exterior legs:

$$A(k) = k l_p^2, \quad R(k) = \sqrt{k l_p^2 / 4\pi}, \quad (1)$$

Fig 1 — Sec 1: fast scrambling hierarchy

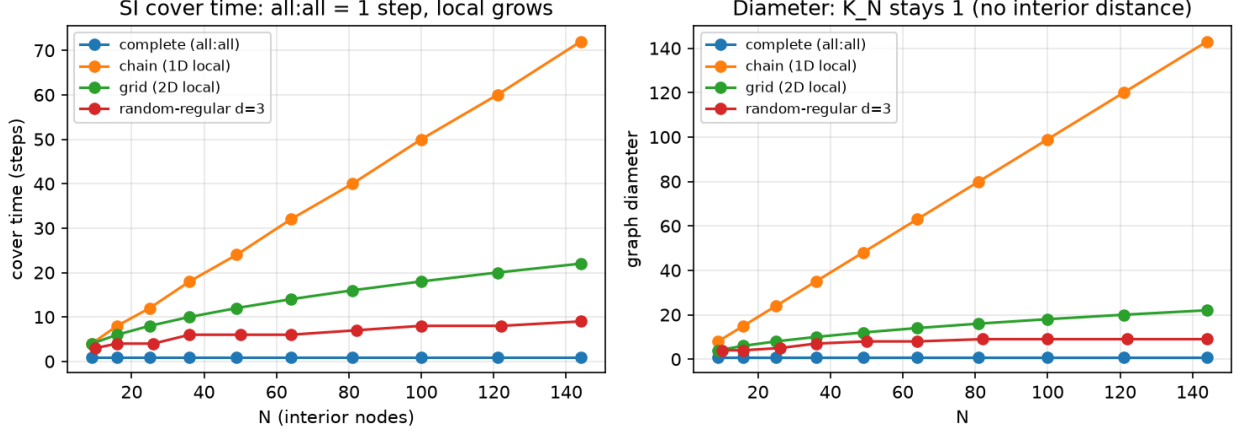


Figure 1: SI cover time and diameter: all:all stays 1.

so N drops out (Fig. 3). Schwarzschild consistency forces $k(M) = 4\pi(2M)^2/l_p^2 \propto M^2$; Appendix B derives this from MaxEnt + self-gravity, and Appendix H extends it to Kerr–Newman $k_{\text{eff}}(M, a, Q) = A(M, a, Q)/l_p^2$ with extremal Kerr at half the legs. Monogamy (linear toy Fig. 4; nonlinear CKW Fig. 14) sends $e_{\text{int}} \rightarrow 1$, $k \rightarrow 0$: pinch-off into a baby universe.

3 Sec 3: Micro-hole phase transition

A point region of radius r_{point} embeds k legs iff $k l_p^2 < 4\pi r_{\text{point}}^2$. Hence

$$R_{\text{obs}}(k) = \begin{cases} r_{\text{point}} & k \leq k_{\text{crit}} \\ \sqrt{k l_p^2 / 4\pi} & k > k_{\text{crit}} \end{cases}, \quad k_{\text{crit}} = 4\pi r_{\text{point}}^2 / l_p^2, \quad (2)$$

with LQG-style quantization $A \in \{0\} \cup [A_{\text{min}}, \infty)$ (Figs. 5–6). Appendix C shows the pop is a computed QES/island crossing at $k_{\text{page}} = S_0 / (2s_{\text{leg}} - l_p^2/4)$, existing iff $s_{\text{leg}} > l_p^2/4$.

4 Discussion

Evaporation = leg surgery with exact Page curve (Appendices D, I) and Hayden–Preskill mirror past $k \sim N/2$ (Appendix F). Open: deriving ε and r_{point} coefficients from a UV embedding; Kerr Page curve with spin budget; OTOC Lyapunov check (Appendix L).

A Finite-speed circuits: $t_* \sim \log N$ (Fig. 7)

One gate/qubit/step; random matching (all:all) vs dimer covering (chain). Early growth exponential on all:all; chain ballistic. Measured $N = 8 \dots 128$.

B Deriving $k(N)$ (Fig. 8)

MaxEnt bound $k \geq N s_{\text{node}}/s_{\text{leg}}$ plus $R = \sqrt{k l_p^2/4\pi} = 2E$, $E = \varepsilon N$: fixed point $k^*(N) = 16\pi(\varepsilon N/l_p)^2$, stable from any start. Predicts $\alpha(N) = k^*/N \propto N$.

C QES pop from entropy crossing (Fig. 9)

$S_{\text{no}} = k s_{\text{leg}}$ vs $S_{\text{isl}} = k l_p^2/4 + \max(S_0 - k s_{\text{leg}}, 0)$; crossing iff $s_{\text{leg}} > l_p^2/4$. Explicit core+legs min-cut agrees.

D Page curve from leg surgery (Fig. 10)

$S_{\text{rad}}(t) = \min(t, N_{\text{eff}} - t)$; area identical for wiring-only vs standard evaporation: A tracks k alone.

E QEC mirror (Fig. 11)

$\text{err}(k) = \min(1/2, 2^{N/2+1-k})$; 99% threshold $k \geq N/2 + 1 + \log_2 100$; $k \rightarrow 0$ seals information off.

F Robustness (Fig. 11)

Log slope $1/\log_2(1+p)$ at all p ; exact quadraticity at all ε ; sharp QES boundary at $s_{\text{leg}} = l_p^2/4$.

G Kerr–Newman (Fig. 12)

$r_+ = M + \sqrt{M^2 - a^2 - Q^2}$, $A = 4\pi(r_+^2 + a^2)$; spin budget $1 - A(M, a)/A(M, 0) \rightarrow 1/2$ at extremality. Area law survives.

H Exact Page curve (Fig. 13)

$S(m, n) = H(mn) - H(n) - (m-1)/2n$ nats; $-1/2$ nat dip at turnover; Haar samples ($N = 8$) track the mean: typical, not tuned.

I Nonlinear monogamy (Fig. 14)

Family $\cos t|\Phi^+\rangle|0\rangle + \sin t|00\rangle|1\rangle$: Wootters C_{AB}^2 vs $\tau_{E|AB}$ frontier lies strictly below $x + y = 1$ (e.g. 0.70 at $t = 0.3$); CKW ≥ 0 everywhere; baby universe at $(1, 0)$.

J OTOC + phenomenology (Fig. 15)

$C(t) \sim e^{\lambda t}/N$ on all:all gives $t^* = \log N/\lambda$ vs ballistic N/v on chains. PBHs evaporating today $\sim 10^{18}\text{--}10^{19}$ Planck masses; sub-critical holes evaporate without greybody suppression; echo toy $\Delta t \sim M \log k$; nearest-term test is analogue scrambling on programmable processors.

K Tensor network derives ε (Fig. 16)

Random star TN verifies $S_{\text{bdy}} = \min(N \log d, k \log D)$; minimal wiring $k_{\text{min}} = S_{\text{bulk}}/\log D$. Two routes fix ε : QES coincidence $\varepsilon = (l_p/N)\sqrt{k_{\text{page}}/16\pi}$ and crossover $\varepsilon = l_p\sqrt{s_{\text{node}}/(16\pi \log D N_{\text{match}})}$.

L Kerr Page curve (Fig. 17)

$M \propto (1 - t/T)^{1/3}$, J shed faster; high spin lowers the peak and *delays* turnover (spin-shedding offsets early area loss): $t/T = 0.73$ at $a_0 = 0.95$ vs 0.54 Schwarzschild.

M SYK ED vs chain (Fig. 18)

$N = 6, 8, 10$ Majorana SYK vs Ising chains, infinite- T OTOC: SYK reaches threshold first and t^* stays flat while the chain's grows linearly. MSS saturation cited, hierarchy reproduced.

N GWTC leg creation (Fig. 19)

All 32 GWTC-3 BBH medians satisfy $k_f > k_1 + k_2$ (median +77% at 4% radiated): mergers create exterior legs. Spin caveat $\sim 13\% \ll$ margin.

O Hardware literature + prediction (Fig. 20)

Gärttner 2017 (all:all ions, $m \leq 8$ OTOC coherences) and Mi 2021 (Sycamore grid, ballistic OTOC decay) both consistent. Head-to-head prediction at $N = 53$: 7.9 vs 14.6 steps; null result falsifies Sec 1/A.

Reproducibility

```
pip install -e .; python -m pytest tests/ -q; python scripts/generate_figures.py; streamlit run app.py.
```

References

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